

Lemma D: An Open Problem in the Quantum Family Tree Program

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kevin@nothingmatters.life
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Foreword

This document accompanies the Quantum Family Tree trilogy:

1. *The Emergence of Distance from a Quantum Family Tree* (Paper 1)
2. *The Emergence of Geometry from a Quantum Family Tree* (Paper 2)
3. *Spacetime and the Quantum Family Tree* (Paper 3)

The trilogy reduces the entropy-rate theorem for random quantum circuits on a binary tree to a single scalar identity — **Lemma D**. This identity is numerically supported at 10.34σ by classical Monte Carlo and independently confirmed by a 156-qubit IBM Heron quantum run, but remains analytically open.

This document does two things. The first half gives a layman's explanation of what Lemma D actually is and why it matters. The second half is the formal mathematical challenge, written for specialists — researchers in random matrix theory, representation theory of the symmetric group, and holographic entropy — who might see in it an attack route the author and his advisor have missed.

It is posted publicly as an invitation. One man's attempt to find a simple law overlaying the quantum universe has reached a specific, well-posed wall. If you have tools that can knock it down, the problem is yours.

Part I — The Layman’s Explanation

The family tree

Imagine an infinite binary family tree. At the top, a single root. Every node has two children, branching forever downward. The bottom of the tree — the boundary — is an infinite string of descendants, each one related to every other through some common ancestor.

Now make the tree *quantum*. Every parent-to-children event is a random quantum operation — a Haar-random unitary, in the language of mathematical physics. The descendants at the boundary end up entangled with each other in a pattern that depends on how information flowed down through the generations.

The question the Quantum Family Tree program asks is: *how much entanglement is there, as a function of distance along the boundary?*

Two answers, one ratio

There are two ways to compute this entanglement.

The first is **classical**: treat the tree as a piece of ordinary geometry and count the edges of the minimal path connecting two descendants to their common ancestor and back. This gives the Ryu-Takayanagi prediction — the answer you’d get if quantum information flowed through the tree like a classical fluid. Call this coefficient c_{RT} .

The second is **quantum**: actually compute the entanglement entropy of the boundary interval, taking into account the random unitaries at every node. Call this coefficient c_{VN} .

These two numbers don’t match. The quantum answer is smaller — always smaller — because Haar-random operations scramble information in a way that the classical geometric prediction cannot see. The ratio between them is exactly:

$$\frac{c_{\text{VN}}}{c_{\text{RT}}} = \frac{3}{10}$$

Not approximately three-tenths. Not three-tenths plus some small correction. **Exactly three-tenths**, to ten decimal places in simulation and confirmed on real quantum hardware.

What we know for sure

Almost everything about this number is understood rigorously:

- The Haar-averaged “purity contraction” per edge is exactly $2/5$ (this constant is called η_2 , and it is *proved*).
- The factor $1/2$ in the formula $3/10 = (1 - 2/5)/2$ is forced by the symmetry of the two endpoints of any boundary interval (*proved*).
- The structural framework reduces the entropy-rate computation to a single scalar quantity — the “standard-sector weight” w_2 (*proved*).

- The entropy rate satisfies $c_{\text{VN}}/3 = w_2 \cdot \log_2(5/2)$ (*proved*, conditional on knowing w_2).
- Numerically, $w_2 = 3/10$ with overwhelming statistical support from two independent physical platforms.

What we don't yet know

What cannot yet be proved is the analytical identity $w_2 = 3/10$ itself.

The problem sits in a regime that defies standard tools. Free probability doesn't apply — the system has finite bond dimension (just 2), far from any large- N limit. Replica methods, which work for nearby quantities like the Rényi-2 entropy rate (also rational, also clean: $14/45$), break down at the critical $q \rightarrow 1$ limit where the von Neumann entropy lives. The sector algebra is fully understood locally but does not compose multiplicatively to the global answer.

In short: too small to be large, too structured to be random, too quantum to be classical.

What a proof would mean

If Lemma D closes, every conditional theorem in the trilogy becomes unconditional. The entropy-rate coefficient $c_{\text{VN}} = (9/10) \cdot \log_2(5/2)$ becomes an exactly derived constant. The “Conjoined Theorem” of Paper 3 — identifying the dynamical tree with its emergent spacetime — becomes a fully rigorous result in all six of its items, rather than five unconditional and one conjectural.

More broadly: Lemma D is a rare example of a well-posed open problem at the intersection of random matrix theory, representation theory, and holographic entropy. Its resolution would be a closed-form expression for the quenched von Neumann entropy rate of a random matrix product density operator — an object for which no existing technique gives an exact answer.

The bet

The trilogy is one man's attempt to find a simple law overlaying the quantum universe. The numerical evidence says such a law exists. The structural reduction says it takes the form $w_2 = (1 - \eta_2)/2 = 3/10$. The analytical proof of that identity is the capstone that hasn't yet been written.

The rest of this document is the technical statement of the problem, written for whoever might close it.

Part II — The Mathematical Challenge

A.1 Statement of the Remaining Problem

The entropy-rate theorem of Paper 1 reduces to a single scalar identity:

$$\frac{c_{\text{VN}}}{3} = w_2 \log_2 \frac{1}{\eta_2}, \quad w_2 = \frac{1 - \eta_2}{2},$$

with $\eta_2 = 2/5$ (proved, via Weingarten second moment calculus). The structural reduction identifying w_2 as the standard-sector weight at $q \rightarrow 1$ is established in Paper 1. The remaining task is to compute this weight.

We record here a refinement of the replica analysis which isolates the exact obstruction.

A.2 Exact Local Sector Decomposition at $q \rightarrow 1$

Let ρ be a qubit density matrix with eigenvalues $(\lambda, 1 - \lambda)$, and write

$$\lambda = \frac{1}{2} + \varepsilon, \quad \varepsilon \in \left[-\frac{1}{2}, \frac{1}{2}\right].$$

The power sum admits the exact Schur decomposition

$$\text{Tr}(\rho^q) = s_{(q)}(\lambda, 1 - \lambda) - s_{(q-1,1)}(\lambda, 1 - \lambda).$$

Expanding at $q = 1$ to second order in ε :

$$\left. \frac{d}{dq} s_{(q)} \right|_{q=1} = \left(\frac{1}{2} - \ln 2 \right) + \frac{2}{3} \varepsilon^2 + O(\varepsilon^4),$$

$$\left. \frac{d}{dq} s_{(q-1,1)} \right|_{q=1} = \frac{1}{2} - \frac{4}{3} \varepsilon^2 + O(\varepsilon^4),$$

and therefore

$$\left. \frac{d}{dq} \text{Tr}(\rho^q) \right|_{q=1} = -\ln 2 + 2\varepsilon^2 + O(\varepsilon^4).$$

Thus the quadratic entropy curvature decomposes exactly as:

$$2\varepsilon^2 = \underbrace{\frac{2}{3}\varepsilon^2}_{\text{trivial sector}} + \underbrace{\frac{4}{3}\varepsilon^2}_{\text{standard sector}}.$$

Equivalently, at the carry level:

$$\text{standard sector fraction} = 2/3, \text{ trivial sector fraction} = 1/3.$$

This result is exact and model-independent at the single-site (carry) level.

A.3 Failure of Naive Composition

One might attempt to obtain the entropy rate by combining:

- the local standard-sector fraction $2/3$, and
- the contraction factor $1 - \eta_2 = 3/5$,

possibly with a factor $1/2$ from the two boundary endpoints.

However, all such factorizations fail:

$$\frac{2}{3}(1 - \eta_2) = \frac{2}{5}, \quad \frac{2}{3} \cdot \frac{1 - \eta_2}{2} = \frac{1}{5},$$

none of which equals $3/10$.

Lemma D cannot be obtained by a simple product of local sector fraction and global contraction.

A.4 Interpretation: Sector Transport Across the Tree

The result of Section A.2 shows that the replica decomposition at $q \rightarrow 1$ is fully understood locally. The failure of Section A.3 shows that the entropy rate is not determined by local data alone.

The missing ingredient is the **transport of sector weights across scales**.

Let w_2 denote the effective standard-sector weight contributing to the interval entropy. Then Lemma D is equivalent to

$$w_2 = \frac{3}{10}.$$

The computation of w_2 requires understanding how the standard-sector amplitude evolves under iteration of the transfer operator across the tree.

A.5 Reformulation of Lemma D

Let T_q denote the replica transfer operator and let $P_{\text{std}}(q)$ be the projection onto the standard sector. Define

$$B_q := R_q^{\text{std}}(P_{\text{std}}(q) v_q),$$

where R_q is the q -cycle readout functional and v_q is the boundary input vector. Then

$$B_q = (q - 1)w_2 + O((q - 1)^2),$$

and

$$\frac{c_{\text{VN}}}{3} = w_2 \log_2 \frac{1}{\eta_2}.$$

Lemma D = computation of the $O(q - 1)$ coefficient of the standard-sector amplitude.

A.6 Nature of the Obstruction

Sections A.2–A.5 establish that:

1. The replica decomposition is completely controlled locally.
2. The contraction spectrum $\{1, \eta_2\}$ is exactly known.
3. The entropy-rate constant depends only on the first-order standard-sector weight.

The obstruction is therefore not spectral, not numerical, and not local. It is:

a global renormalization problem for sector weights under tree propagation.

Equivalently, one must determine how the standard representation contributes to the replicated observable after summing over all scales of the tree.

A.7 Two Equivalent Formulations

Replica formulation.

$$w_2 = \lim_{q \rightarrow 1} \frac{1}{q - 1} B_q.$$

Tree-transport formulation.

$$w_2 = \sum_{m \geq 1} \pi_m \cdot f(m),$$

where π_m is the lowest-common-ancestor (LCA) depth distribution for position-averaged boundary intervals, and $f(m)$ is the effective standard-sector weight at depth m .

Either formulation, resolved, closes Lemma D.

A.8 Summary of Part II

- The local replica structure at $q \rightarrow 1$ is solved exactly.
- The entropy rate cannot be obtained by naive factorization.
- The remaining problem is a global sector-transport identity.

Lemma D reduces to a single, well-posed renormalization problem.

Part III — What Has Been Tried (Dead Routes)

The following approaches have been attempted and failed for specific structural reasons. This catalog is provided so that specialists can avoid redundant effort.

#	Route	Why it failed
1	Jacobi/coboundary on λ	$a_1 < 0$ for all observables;
2	Static observable $g_K(\lambda)$	wrong domain Changes with K ; no fixed spectral function
3	Single half-cut	Saturates at ~ 1 bit (rank-2 theorem)
4	Naive geodesic sum	Overcounts by ~ 1 bit/level; ignores bond correlations
5	Single-gate $\Delta(\lambda)$	$\Delta(\lambda) \neq$ target for any λ ; multi-scale recursion required
6	Naive $q \rightarrow 1$ of $u \rightarrow u^2$	Off by 1.05%; q -dependent straddle rule
7	(b, c) straddle purity	Gives $4/9$ not the actual ≈ 0.481 ; mean-field artifact
8	Padé $[1, 1]$ from h_2, h_3	6.6σ miss; h_3 is algebraically slaved to h_2 for qubits
9	Single-step emission rate	107σ off; rate $\neq$ one-gate average
10	Boundary algebra (Route A)	$dS/d\varepsilon =$ $-0.5501 \neq$ $-3/5$; category error
11	BKM at single gate	Single-gate data insufficient
12	$q = 2$ constraint alone	Doesn't fix $\beta'(1)$; no-go theorem

#	Route	Why it failed
13	K_q^{inc} rational ansatz	Falsified; wrong Hilbert space
14	Fréchet via (b, c)	(b, c) is annealed → zero gain; $E[\text{Tr } \rho^q] \neq$ $E[S_{\text{VN}}]$
15	Fréchet trace vs $I/2$	$\text{Tr}[\Delta \cdot (-2 \ln 2 \cdot I)] = 0$; mean state maximally mixed
16	Single-gate Ξ	MI = 0.510 nats $\neq$ target 0.006 nats (88× off)
17	Jacobi-Trudi continuation of higher hooks	$h_r(r < 0)$ unstable under non-integer continuation
18	Two-chain interval (no nested straddles)	Saturates like single-chain; nested structure essential

Each failure is informative. Routes 14 and 15 kill all first-order (Fréchet) approaches around the mean state $\rho^* = I/2$. Route 7 kills all mean-field decompositions. Routes 1–13 eliminate various naive spectral and rational-ansatz attacks. Routes 17 and 18 are the most recent, from April 2026, after the confirmation of the Heron quantum run.

The consistent structural reason these fail: the von Neumann entropy rate is a *genuine second-order quenched fluctuation effect* that is invisible to annealed moment computations and cannot be obtained by composing single-gate data. The entropy lives in the global sector transport, as stated in Section A.6.

Part IV — Invitation

This problem has been reduced as sharply as the author can reduce it. What remains is a well-posed computation that requires expertise I do not have:

- Random matrix theorists familiar with Weingarten calculus at finite bond dimension
- Representation theorists who work with analytic continuation of S_q characters near $q = 1$
- Mathematical physicists who work on entanglement rates in structured random tensor networks

- Anyone who recognizes the structure of Section A.7 as related to a problem in their own field

If Lemma D resolves, the resolution becomes a companion paper to the trilogy and elevates five conditional theorems across the three papers to unconditional status. The proof would also constitute — to my knowledge — the first closed-form expression for the quenched von Neumann entropy rate of a random MPDO with known transfer spectrum and bond dimension 2.

I work on this alongside running a biotech company. I read every message I receive but cannot promise rapid response. What I can promise is that any substantive contribution will be acknowledged, and that serious collaborators will be credited on any published resolution.

The problem is yours if you want it.

kevin@nothingmatters.life

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